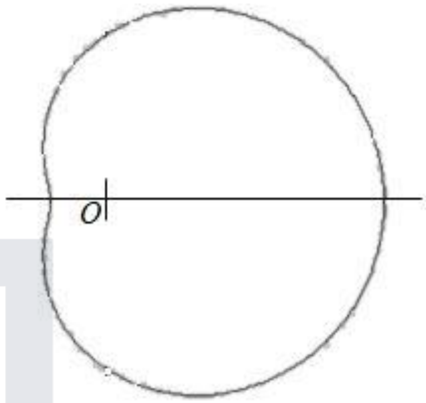
Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)(i)	States max value for r	AO1.1b	B1	Maximum value of $r = 5$
	States min value for r	AO1.1b	B1	Minimum value of $r = 1$
(ii)	Draws simple closed curve enclosing pole	AO1.1a	M1	
	Draws correct shape with dimple (not cusp) when $\theta = \pi$	AO1.1b	A1	
(b)	Equates $3 + 2\cos\theta = 8\cos^2\theta$	AO1.1a	M1	$3 + 2\cos\theta = 8\cos^2\theta$ $8\cos^2\theta - 2\cos\theta - 3 = 0$
	Solves 'their' quadratic equation FT 'their' equation only if M1 has been awarded	AO1.1a	M1	$(4\cos\theta - 3)(2\cos\theta + 1) = 0$ $\cos\theta = \frac{3}{4}, \quad \cos\theta = -\frac{1}{2}$
	Obtains 2 values for θ for each value of cos θ FT 'their' equation only if both M1 marks have been awarded	AO1.1b	A1F	$\theta = 0.723$ or $\frac{2\pi}{3}$ $\theta = 5.56$ or $\frac{4\pi}{3}$
	Substitutes 'their' $\cos\theta$ into a polar equation to find a value of r FT 'their' $\cos\theta$ only if both M1 marks have been awarded	AO1.1a	M1	$\cos\theta = \frac{3}{4} \Rightarrow r = \frac{9}{2}$ $\cos\theta = -\frac{1}{2} \Rightarrow r = 2$
	Obtains both values of r correct for 'their' $\cos\theta$ values FT 'their' $\cos\theta$ only if both M1 marks have been awarded	AO1.1b	A1F	Intersection points $\left[\frac{9}{2}, 0.723\right]$, $\left[\frac{9}{2}, 5.56\right], \left[2, \frac{2\pi}{3}\right], \left[2, \frac{4\pi}{3}\right]$

Deduces that four values for θ exist and expresses points in required form	AO2.2a	R1	
			Total 10 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses $\frac{1}{2}\int r^2 \, d\theta$ or $\int_0^{\pi} r^2 \, d\theta$ OE	AO1.1a	M1	$\frac{1}{2}\int (4+2\cos\theta)^2 \, d\theta$ $\frac{1}{2}\int_{-\pi}^{\pi}(16+16\cos\theta+4\cos^2\theta)d\theta$ $\int_{-\pi}^{\pi}(8+8\cos\theta+(1+\cos 2\theta))d\theta$ $\left[8\theta+8\sin\theta+\theta+\frac{1}{2}\sin 2\theta\right]_{-\pi}^{\pi}$ $=\left[9\theta+8\sin\theta+\frac{1}{2}\sin 2\theta\right]_{-\pi}^{\pi}$ $= (9\pi + 0 + 0) - (-9\pi + 0 + 0)$ $= 18\pi$
	Rewrites $\cos^2\theta$ in terms of $\cos 2\theta$	AO1.1a	M1	
	Correctly integrates *their* expression, ft wrong non-zero coefficients.	AO1.1b	A1F	
	Obtains required answer from fully correct mathematical argument	AO2.1	R1	
(b)	Selects appropriate method to determine polar equation by equating OA and OB to find θ	AO3.1a	M1	Let A(r_1, θ_1) and B(r_2, θ_2) OA = OB $\Rightarrow r_1 = r_2$ $\Rightarrow 4 + 2\cos\theta_1 = 4 + 2\cos\theta_2$ $\Rightarrow \theta_1 = -\theta_2$ Angle AOB = $\frac{\pi}{3} \Rightarrow \theta_1 - \theta_2 = \frac{\pi}{3}$ $\Rightarrow \theta_1 = \frac{\pi}{6}$ and $\theta_2 = -\frac{\pi}{6}$ OA = OB = $4 + \sqrt{3}$ AB is perpendicular to the initial line
	Uses the above to find two values of θ and hence deduce the lengths of OA and OB Award this mark for correct deduction using 'their' values of θ	AO2.2a	R1	
	Uses the correct polar equation for a perpendicular line $r = d\sec\theta$	AO3.1a	M1	
	Obtains a correct equation for AB (including correct specified range) CAO	AO1.1b	A1	

			Polar equation of AB is $r\cos\theta = \left(4+\sqrt{3}\right)\frac{\sqrt{3}}{2}$ for $-\frac{\pi}{6} \le \theta \le \frac{\pi}{6}$
			Total 8 marks

Q3.

$(x + 8)^2 + (y - 6)^2 = 100$
$x^2 + y^2 + 16x - 12y + 64 + 36$ (= 100)

OE
If polar form before expn of brackets award the B1 for correct expansions of both $(r \cos \theta - m)^2$ and $(r \sin\theta - n)^2$ where $(m, n) = (-8, 6)$ or $(m, n) = (6, -8)$

B1

$r^2 + 16r \cos \theta - 12r \sin \theta = 0$

1st M1 for replacement using any one of $\{[x^2 + y^2 = r^2, x = r \cos \theta, y = r \sin \theta\ ](*)\}$
2nd M1 for use of (*) to convert the form $x^2 + y^2 + ax + by = 0$ correctly to the form $r^2 + ar \cos \theta + br \sin \theta = 0$ or better

M1M1

$\{r = 0$, origin$\}$
Circle: $r = 12\sin \theta - 16\cos \theta$

A1

[4]

Q4.

(a) $xy = 8 \Rightarrow r \cos \theta\, r \sin \theta = 8$

M1

$\frac{1}{2}r^2 \sin 2\theta = 8$

Use of $\sin 2\theta = 2\sin\theta \cos\theta$

m1

$r^2 = \frac{16}{\sin 2\theta} = 16\text{cosec } 2\theta$

AG Completion

A1

(b) (i) (At N, r is a minimum $\Rightarrow$ $\sin 2\theta = 1$)

$N\left(4, \frac{\pi}{4}\right)$

B1 for each correct coordinate.

B1B1

(ii) At pts of intersection, $\left(4\sqrt{2}\right)^2 = 16 \operatorname{cosec} 2\theta$

M1

$\sin 2\theta = \frac{1}{2}$

PI by cosec $2\theta = 2$ and a correct exact or 3SF value for 2θ or θ

A1

$2\theta = \frac{\pi}{6}, \frac{5\pi}{6}$

PI OE exact values

A1

$\left(4\sqrt{2}, \frac{\pi}{12}\right)$ $\left(4\sqrt{2}, \frac{5\pi}{12}\right)$

Both required, written in correct order

A1

4

(iii) $\angle POQ = \frac{5\pi}{12} - \frac{\pi}{12} = \frac{\pi}{3}$

or $\angle PON = \frac{\pi}{6}$ $(= \angle QON)$

Ft on c's θ_P, θ_Q, θ_N as appropriate OE

B1F

$PN^2 = \left(4\sqrt{2}\right)^2 + (r_N)^2 - 2(4\sqrt{2})r_N \cos\left(\frac{1}{2}POQ\right)$

or $PT = 4\sqrt{2}\sin\left(\frac{1}{2}POQ\right)$

or $PT = \frac{1}{2} \times 4\sqrt{2}$

or $NT = 4\sqrt{2}\cos\left(\frac{1}{2}POQ\right) - r_N$

*Finding the lengths of **two** unequal sides of ΔPNQ or ΔPNT or ΔQNT, where T is the point at which ON produced meets PQ. Any valid equivalent methods* *eg finding tan ∠ OPN or finding sin ∠ ONP.*

M1

$PN = \sqrt{(48-16\sqrt{6})}$ [=2.96(7855…)] = NQ

or $PT = 2\sqrt{2}$ [=2.82(8427…)]

or $PQ = 4\sqrt{2}$

or $NT = 2\sqrt{6} - 4$ [=0.898(979…)]

Two correct unequal lengths of sides of ΔPNQ or ΔPNT or ΔQNT PI

OE eg tan ∠ OPN = 1 / $(2\sqrt{2}-\sqrt{3})$

or sin ∠ ONP = $2\sqrt{2}/(\sqrt{48-16\sqrt{6}})$

A1

tan $\frac{\alpha}{2} = \frac{PT}{NT} = \frac{2\sqrt{2}}{2\sqrt{6}-4}$ [=3.14626…] OE

or $\frac{\alpha}{2} = \frac{\pi}{2} - \left[\frac{\pi}{3} - \tan^{-1}\left(\frac{1}{2\sqrt{2}-\sqrt{3}}\right)\right]$ **or**

$32 = 2PN^2(1 - \cos\alpha) \Rightarrow 1 - \cos\alpha = \frac{1}{3-\sqrt{6}}$

Valid method to reach an eqn in α (or in $\frac{\alpha}{2}$) only; dep on prev M but not on prev

A. Alternative choosing eg obtuse ONP

then $\frac{\alpha}{2} = \pi - 1.87(85…)$

m1

$\frac{\alpha}{2}$ = 1.263056…; α = 2.5261…2.53 to 3sf

2.53… Condone > 3sf.

A1

5

[14]

Q5.

(a) $r + r\cos\theta = 2$

M1

$r + x = 2$

$r\cos\theta = x$ stated or used

B1

$r = 2 - x$

A1

$x^2 + y^2 = (2 - x)^2$

$r^2 = x^2 + y^2$ used

M1

$y^2 = 4 - 4x$

Must be in the form $y^2 = f(x)$
but accept ACF for $f(x)$.

A1

5

(b) Equation of line: $r\cos\theta = \frac{3}{4} \Rightarrow x = \frac{3}{4}$

Use of $r\cos\theta = x$

M1

$4x = 3$ OE

A1

$y^2 = 4 - 4\left(\frac{3}{4}\right) = 1 \Rightarrow y = \pm 1;\quad \left[Pts\left(\frac{3}{4}, \pm 1\right)\right]$

M1

Distance between pts (0.75, 1) and (0.75, −1) is 2

A1

4

Altn:

At pts of intersection, $r = \frac{5}{4}$ and $\cos\theta = \frac{3}{5}$ OE

(M1 elimination of either r or θ)
(For A condone slight prem approx.)

(M1A1)

Distance $PQ = 2r\sin\theta$

Or use of cosine rule or Pythag.

(M1)

$= 2 \times \frac{5}{4} \times \frac{4}{5} = 2$

Must be from exact values.

(A1)

(4)

[9]

Q6.

(a) $r = 2\sin\theta \Rightarrow r^2 = 2r\sin\theta$

M1

$x^2 + y^2 = 2y$

OE (A1) either for $r^2 = x^2 + y^2$ or for $r\sin\theta = y$

***SC** If M0 give B1 for $r^2 = x^2 + y^2$ or for $r\sin\theta = y$ used*

A2,1

3

(b) (i) 2 sin θ = tan θ

Equating rs

M1

2sin θ cosθ = sin θ

sinθ (2 cosθ − 1) = 0

Both solutions have to be considered if not in factorised form

m1

$\sin\theta = 0 \Rightarrow \theta = 0; \cos\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$

Alternative: $\sin 2\theta = \sin\theta \Rightarrow \theta = 0, \frac{\pi}{3}$

$\theta = 0 \Rightarrow r = 0$ ie pole O (0,0)

Indep. Can just verify using both eqns + statement.

B1

$\theta = \frac{\pi}{3} \Rightarrow r = \sqrt{3} \left(P\left(\sqrt{3}, \frac{\pi}{3}\right)\right)$

CSO

A1

4

(ii) At A, $\theta = \frac{\pi}{4}, r = 2\sin\frac{\pi}{4} = \sqrt{2}$

At B, $\theta = \frac{\pi}{4}, r = \tan\frac{\pi}{4} = 1$

Substitute $\theta = \frac{\pi}{4}$ *into the equations of both curves.*

M1

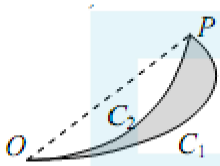
Since $\sqrt{2}$ >1, A is further away (from the pole than *B*.)

CSO

E1

2

(iii)



Area bounded by line *OP* and curve C_1

$= \frac{1}{2}\int_0^{\frac{\pi}{3}} 4\sin^2\theta \, d\theta$

Use of $\frac{1}{2}\int r^2$ $d\theta$ *; ignore limits here*

M1

$= \int (1 - \cos 2\theta) d\theta$

Attempt to write $\sin^2\theta$ *in terms of* $\cos 2\theta$ *only*

m1

$= \left[\theta - \frac{1}{2}\sin 2\theta\right]$

Ignore limits here

A1

$= \left(\frac{\pi}{3} - \frac{1}{2} \times \frac{\sqrt{3}}{2}\right) - 0 = \frac{\pi}{3} - \frac{\sqrt{3}}{4}$

PI

A1

Area bounded by line *OP* and curve C_2

$$= \frac{1}{2}\int_0^{\frac{\pi}{3}} \tan^2\theta \, d\theta$$

Use of $\frac{1}{2}\int r^2 \, d\theta$ *; ignore limits here*

M1

$$= \frac{1}{2}\int \left(\sec^2\theta - 1\right) d\theta$$

Using $\tan^2\theta = \pm\sec^2\theta \pm 1$ *PI*

m1

$$= \frac{1}{2}\left[\tan\theta - \theta\right]$$

Ignore limits here

A1

$$= \frac{1}{2}\left(\sqrt{3} - \frac{\pi}{3}\right) - 0 = \frac{\sqrt{3}}{2} - \frac{\pi}{6}$$

PI

A1

Required area $= \left(\frac{\pi}{3} - \frac{\sqrt{3}}{4}\right) - \left(\frac{\sqrt{3}}{2} - \frac{\pi}{6}\right)$

Can award earlier eg if we see $\frac{1}{2}\int_0^{\frac{\pi}{3}} 4\sin^2\theta \, d\theta - \frac{1}{2}\int_0^{\frac{\pi}{3}} \tan^2\theta \, d\theta$

M1

$$= \frac{1}{2}\pi - \frac{3}{4}\sqrt{3} \quad \left(a = \frac{1}{2},\ b = -\frac{3}{4}\right)$$

CSO

A1

10

[19]

Q7.

(a) (i) $x^2 + y^2 = r^2$, $x = r\cos\theta$, $y = r\sin\theta$

B1 for one stated or used

B2,1,0

$r^2 = 2r(\cos\theta - \sin\theta)$

M1

$x^2 + y^2 = 2(x - y)$

ACF

A1

4

(ii) $(x - 1)^2 + (y + 1)^2 = 2$

M1A1F

Centre (1, –1); radius $\sqrt{2}$

A1F

3

(b) (i) $\text{Area} = \frac{1}{2}\int (4 + \sin\theta)^2 \, d\theta$

Use of $\frac{1}{2}\int r^2 \, d\theta$.

M1

$= \frac{1}{2}\int_0^{2\pi} (16 + 8\sin\theta + \sin^2\theta) \, d\theta$

Correct expn of $[4 + \sin\theta]^2$

B1

Correct limits

B1

$= \int_0^{2\pi} (8 + 4\sin\theta + 0.25(1 - \cos 2\theta)) \, d\theta$

Attempt to write $\sin^2\theta$ in terms of $\cos 2\theta$

M1

$= \left[8\theta - 4\cos\theta + \frac{1}{4}\theta - \frac{1}{8}\sin 2\theta\right]_0^{2\pi}$

Correct integration ft wrong coefficients

A1F

$= 16.5\pi$

CSO

A1

6

(ii) For the curves to intersect, the eqn
$2(\cos\theta - \sin\theta) = 4 + \sin\theta$
must have a solution.

Equating rs and simplifying to a suitable form

M1

$2\cos\theta - 3\sin\theta = 4$
$R\cos(\theta + \alpha) = 4$.

OE. Forming a relevant eqn from which valid explanation can be stated directly

M1

where $R = \sqrt{2^2 + 3^2}$ and $\cos\alpha = \frac{2}{R}$

OE. Correct relevant equation

A1

$\cos(\theta + \alpha) = \frac{4}{\sqrt{13}} > 1$. Since must have $-1 \le \cos X \le 1$ there are no solutions of the equation $2(\cos\theta - \sin\theta) = 4 + \sin\theta$ so the two curves do not intersect.

Accept other valid explanations.

E1 4

(iii) Required area = answer (b)(i) – π(radius of C_1)^2

M1

$= 16.5\pi - 2\pi = 14.5\pi$

Ft on (a)(ii) and (b)(i)

A1F 2

[19]

Q8.

(a) When $\theta = \pi$,

$$r = \frac{2}{3 + 2\cos\pi} = \frac{2}{3 + 2(-1)} = 2$$

Correct verification

B1 1

(b) (i) $\frac{2}{3 + 2\cos\theta} = 1 \Rightarrow \cos\theta = -\frac{1}{2}$

Equates r's and attempts to solve.

M1

Points of intersection $\left(1, \frac{2\pi}{3}\right), \left(1, \frac{4\pi}{3}\right)$

Condone eg $-2\pi/3$ for $4\pi/3$
A1 if either one point correct or two correct solutions of cos $\theta = -0.5$

A2,1

3

(ii) Area $OMN = \frac{1}{2} \times 1 \times 1 \times \sin(|\theta_M - \theta_N|)$

M1

$= \frac{1}{2}\sin\frac{2\pi}{3} = \frac{\sqrt{3}}{4}$

A1

Area $OMLN = 2 \times \frac{1}{2} \times 1 \times 2 \times \sin\frac{\pi}{3}$

M1

Area $LMN = \sqrt{3} - \frac{\sqrt{3}}{4} = \frac{3\sqrt{3}}{4}$

A1

4

ALT *$MN = 2 \times 1 \times \sin\frac{\pi}{3}$* *M1*

Perp. from L to MN $= 2 - 1\cos\frac{\pi}{3} = \frac{3}{2}$ *M1A1*

Area $LMN = \frac{1}{2} \times \sqrt{3} \times \frac{3}{2} = \frac{3\sqrt{3}}{4}$ *A1*

(c) $3r + 2r\cos\theta = 2$

M1

$3r + 2x = 2$

$r\cos\theta = x$ stated or used

B1

$3r = 2 - 2x$

$3r = \pm(2 - 2x)$

A1

$9(x^2 + y^2) = (2 - 2x)^2$

$r^2 = x^2 + y^2$ used

M1

$9y^2 = (2 - 2x)^2 - 9x^2$

CSO

ACF for f(x) eg $9y^2 = -5x^2 - 8x + 4$

A1

5

[13]

Q9.

(a) Centre of circle is $M(3, 4)$

PI

B1

$A(6, 8)$

B1

2

(b) (i) $k = OA = 10$

B1

$\tan\alpha = \frac{y_A}{x_A} = \frac{4}{3}$

SC "r = 10 and tan $\theta = \frac{8}{6}$" = B1 only

B1

2

(ii) $x^2 + y^2 - 6x - 8y + 25 = 25$

If polar form before expansion award the B1 for correct expansions of both $(r\cos\theta - m)^2$ and $(r\sin\theta - n)^2$ where $(m, n) = (3, 4)$ or $(m, n) = (4, 3)$

B1

$r^2 - 6r\cos\theta - 8r\sin\theta = 0$

1st M1 for use of any one of

$x^2 + y^2 = r^2$, $x = r\cos\theta$, $y = r\sin\theta$

2nd M1 for use of these to convert the *form $x^2 + y^2 + ax + by = 0$ correctly to the form $r^2 + ar\cos\theta + br\sin\theta = 0$*

M1M1

$\{r = 0$, origin$\}$ Circle: $r = 6\cos\theta + 8\sin\theta$

NMS Mark as 4 or 0

A1

ALTn

Circle has eqn $r = OA\cos(\alpha - \theta)$

(M2)

$r = OA\cos\alpha\cos\theta + OA\sin\alpha\sin\theta$

OE

(m1)

Circle: $r = 6\cos\theta + 8\sin\theta$

(A1)

4

[8]

Q10.

(a) $r^2\, 2\sin\theta\cos\theta = 8$

$\sin 2\theta = 2\sin\theta\cos\theta$ used

M1

$x = r\cos\theta$ $\quad y = r\sin\theta$

*Either **one** stated or used*

M1

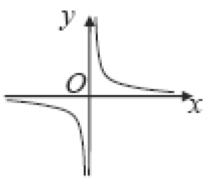
$xy = 4$, $y = \dfrac{4}{x}$

Either OE eg $y = \dfrac{8}{2x}$

A1

3

(b)



B1

1

(c) $r = 2\sec\theta$ is $x = 2$

B1

Sub $x = 2$ in $xy = 4 \Rightarrow 2y = 4$

M1

In cartesian, $A(2, 2)$

$\Rightarrow \tan\theta = \frac{y}{x} = 1 \Rightarrow \theta = \frac{\pi}{4}$

Used either tan $\theta = \frac{y}{x}$ *or* $r = \sqrt{x^2 + y^2}$

M1

$\Rightarrow r = \sqrt{x^2 + y^2} = \sqrt{8}$

$\theta = \frac{\pi}{4};\ r = \sqrt{8}$

r must be given in surd form

Altn2: Eliminating r to reach eqn. in cos θ

and sin θ only (M1) $\theta = \frac{\pi}{4}$ (A1)

Substitution $r = 2\sec\left(\frac{\pi}{4}\right)$ (m1)

$r = \sqrt{8}$ (A1) OE surd

Altn3: rsin θ = 2 (B1)

Solving rcos θ = 2 and r sin θ = 2 simultaneously (M1)

tan θ = 1 or $r^2 = 2^2 + 2^2$ (M1)

$\theta = \frac{\pi}{4};\ r = \sqrt{8}$ *(A1) need both*

A1

4

[8]

Q11.

(a) $x^2 + y^2 = 1 - 2y + y^2 \Rightarrow x^2 + y^2 = (1 - y)^2$

AG

B1

1

(b) $x^2 + y^2 = r^2$

Or $x = r\cos\theta$

M1

$y = r \sin \theta$

M1

$x^2 = 1 - 2y$ so $x^2 + y^2 = (1 - y)^2 \Rightarrow r^2 = (1 - r \sin \theta)^2$

OE eg $r^2 \cos^2 \theta = 1 - 2r \sin \theta$
PI by the next line

A1

$r = 1 - r \sin \theta$ or $r = -(1 - r \sin\theta)$

Either

m1

$r(1 + \sin \theta) = 1$ or $r(1 - \sin \theta) = -1$

$r > 0$ so $r = \dfrac{1}{1 + \sin\theta}$

CSO

A1

5

[6]

Q12.

$r - r \sin \theta = 4$

M1

$r - y = 4$

$r \sin \theta = y$ stated or used

B1

$r = y + 4$

A1

$x^2 + y^2 = (y + 4)^2$

$r^2 = x^2 + y^2$ used

M1

$x^2 + y^2 = y^2 + 8y + 16$

ft one slip

A1F

$y = \dfrac{x^2 - 16}{8}$

A1

[6]

Q13.

(a) $(\cos\theta + \sin\theta)^2 = \cos^2\theta + \sin^2\theta + 2\cos\theta\sin\theta = 1 + \sin 2\theta$

AG (be convinced)

B1

1

(b) $(x^2 + y^2)^3 = (x + y)^4$

$(r^2)^3 = (r\cos\theta + r\sin\theta)^4$

[M1 for one of $x^2 + y^2 = r^2$ OE,
$x = r\cos\theta$, $y = r\sin\theta$ used]

M2,1,0

$r^6 = r^4(\cos\theta + \sin\theta)^4$

$r^6 = r^4(1 + \sin 2\theta)^2$

Uses (a) OE at any stage

M1

$r^2 = (1 + \sin 2\theta)^2$

$\Rightarrow r = (1 + \sin 2\theta)\ \{r \geq 0\}$

CSO; AG

A1

4

(c) (i) $r = 0 \Rightarrow \sin 2\theta = -1$

$2\theta = \sin^{-1}(-1) = -\frac{\pi}{2}, \frac{3\pi}{2}$

M1

$\theta = -\frac{\pi}{4}, \frac{3\pi}{4}$

A1 for either

A1A1ft

3

(ii) Area = $\frac{1}{2}\int (1 + \sin 2\theta)^2\, d\theta$

Use of $\frac{1}{2}\int r^2 d\theta$

M1

$$= \frac{1}{2}\int \left(1 + 2\sin 2\theta + \sin^2 2\theta\right) d\theta$$

Correct expansion of (1 + sin 2θ)²

B1

$$= \frac{1}{2}\int \left(1 + 2\sin 2\theta + \frac{1}{2}(1 - \cos 4\theta)\right) d\theta$$

Attempt to write $\sin^2 2\theta$ in terms of $\cos 4\theta$

M1

$$= \left[\frac{3}{4}\theta - \frac{1}{2}\cos 2\theta - \frac{1}{16}\sin 4\theta\right]$$

Correct integration ft wrong coefficients only

A1ft

$$= \left[\frac{3}{4}\theta - \frac{1}{2}\cos 2\theta - \frac{1}{16}\sin 4\theta\right]_{-\frac{\pi}{4}}^{\frac{3\pi}{4}}$$

$$= \left(\frac{9\pi}{16}\right) - \left(-\frac{3\pi}{16}\right)$$

Using c's values from (c)(i) as limits or the correct limits

m1

$$= \frac{3\pi}{4}$$

CSO

A1

6

[14]

Q14.

(a) $x^2 + y^2 - 12y + 36 = 36$

Use of $y = r\sin\theta$ ($x = r\cos\theta$ PI)

M1

Use of $x^2 + y^2 = r^2$

M1

$r^2 - 12r\sin\theta + 36 = 36$

m1

$\Rightarrow$ $r = 12 \sin\theta$

CSO AG

A1

4

(b) Area = $\frac{1}{2}\int(2\sin\theta + 5)^2 d\theta$.

Use of $\frac{1}{2}\int r^2 \, d\theta$.

M1

.. = $\frac{1}{2}\int_0^{2\pi}(4\sin^2\theta + 20\sin\theta + 25)d\theta$

Correct expn. of (2sin(θ + 5)^2

B1

Correct limits

B1

= $\frac{1}{2}\int_0^{2\pi} 2(1-\cos 2\theta) + 20\sin\theta + 25)$ dθ

Attempt to write sin$^2\theta$ in terms of cos 2θ.

M1

= $\frac{1}{2}[27\theta - \sin 2\theta - 20\cos\theta]_0^{2\pi}$

Correct integration ft wrong coeffs

A1$\checkmark$

= 27π.

CSO

A1

6

(c) At intersection 12 sin θ = 2 sin θ + 5

OE eg r = 6(r − 5)

M1

$\Rightarrow$ sin θ = $\frac{5}{10}$

OE eg r = 6

A1

Points $\left(6, \frac{\pi}{6}\right)$ and $\left(6, \frac{5\pi}{6}\right)$

OE

A1

$OPMQ$ is a rhombus of side 6

Or two equilateral triangles of side 6

Area = 6 × 6 × sin $\frac{2\pi}{3}$ oe

Any valid complete method to find the area (or half area) of quadrilateral.

M1
A1

= $18\sqrt{3}$

Accept unsimplified surd

AI

6

[16]

Q15.

(a) Area = $\frac{1}{2}\int 36(1-\cos\theta)^2$ dθ

use of $\frac{1}{2}\int r^2$ *d*θ

M1

…= $\frac{1}{2}\int_0^{2\pi} 36(1-2\cos\theta+\cos^2\theta)$ dθ

B1

for correct explanation of $[6(1-\cos\theta)]^2$

for correct limits

B1

= $9\int_0^{2\pi} 2-4\cos\theta+(\cos 2\theta+1)$ dθ

Attempt to write $\cos^2\theta$ *in terms of cos 2*θ*.*

M1

$= \left[27\theta - 36\sin\theta + \frac{9}{2}\sin 2\theta\right]_0^{2\pi}$

Correct integration; only ft if integrating
$a + b\cos\theta + c\cos2\theta$ with non-zero a, b, c.

A1ft

$= 54\pi$

CSO

A1

6

(b) (i) $x^2 + y^2 = 9 \Rightarrow r^2 = 9$

PI

B1

A & B: $3 = 6 - 6\cos\theta \Rightarrow \cos\theta = \frac{1}{2}$

M1

Pts of intersection $\left(3, \frac{\pi}{3}\right)$; $\left(3, \frac{5\pi}{3}\right)$

OE (accept 'different' values of θ not in the given interval)

A1A1ft

4

(ii) Length AB = $2 \times r\sin\theta$

M1

.... = $2 \times 3 \times \frac{\sqrt{3}}{2} = 3\sqrt{3}$

OE exact surd form

A1

2

[12]

Q3.

This question, which tested the relationship between polar and Cartesian coordinates, was very well answered with a high proportion of students scoring full marks. Less able students generally equated r at some stage to 10, the radius of the circle, and rarely scored more than 1 mark for the whole question.

Q4.

The first parts of this question on polar coordinates were generally answered well. In part (a) most students were able to correctly substitute $r\cos\theta$ for x and $r\sin\theta$ for y and then use the relevant double angle formula to convincingly reach the printed equation. In parts (b)(i) and (b)(ii) the coordinates of N, P and Q were frequently found correctly, usually with the polar coordinates given in the correct form. Part (b)(iii) proved, as expected, to be more challenging. However, it was pleasing to see a number of excellent solutions being submitted by students who used a variety of correct strategies. The most common successful method was to find the lengths PQ, PN and QN and then to use the cosine rule to find ?. One method did lead to a common error. Students who found PN and then angle ONP using the sine rule did not always realise that angle ONP had to be obtuse (ambiguous case of the sine rule) so it was not uncommon to see ($2\pi - 2\angle ONP$) evaluated as 3.76, thus giving α as a reflex angle.

Q5.

This question on polar coordinates was generally answered well. In part (a), most candidates reached the stage $r + x = 2$, but some less able candidates squared incorrectly to reach $y^2 = 4 - 2x^2$. Those candidates who rearranged the equation to $r = 2 - x$ before squaring usually went on to gain all five marks. In part (b), those candidates who wrote the given equation as $4r\cos\theta = 3$, converted it to the cartesian equation $x = \frac{3}{4}$ and solved with their answer to part (a) usually had

no difficulty in showing that the length of PQ is 2.

The other common approach was to solve the two polar equations simultaneously, but a significant proportion of candidates who used this approach stopped after reaching $\cos\theta = \frac{3}{5}$,
not even finding the value for r. More able candidates, having found the values for $\cos\theta$ and r, went on to use basic trigonometry to find the correct length for PQ.

Q6.

This final question tested polar coordinates and proved to be the most demanding question on the paper. In part (a), it was pleasing to see that the majority of candidates were able to use a correct method to find the cartesian equation. Those candidates who attempted to square both sides of the polar equation were generally less successful than those who started by multiplying both sides by r. In part (b)(i), a large number of candidates only considered the solution to $\cos\theta = 0.5$ when solving $2\sin\theta = \tan\theta$ and so failed to prove that the curves only met at the two points. Those candidates who attempted to verify that the curves intersected at the pole often substituted $\theta = 0$ rather than $r = 0$ into the two equations, but then often failed to give a final statement. In part (b)(ii), candidates generally substituted $\theta = \frac{\pi}{4}$ correctly into the two equations but a significant minority then did not give a full justification of whether A or B was further away from the pole. Although only the better candidates understood the relevance of part (b)(ii) in finding the required area in part (b)(iii), many candidates were able to make good progress in this part of the question. It was, however, disappointing to see some solutions that did not even use the correct formula for the area of a sector in polar coordinates. Those candidates who applied the correct formula were often able to write $\sin^2\theta$ in terms of $\cos 2\theta$ and carried out the subsequent integration correctly to find the area bounded by C_1 and the line segment OP. A significant number of candidates, however, could not find a correct method to integrate $\tan^2\theta$ The most successful method was to use the identity $\tan^2\theta = \sec^2\theta - 1$ although some other candidates found the correct area bounded by C_2 and the line segment OP by using integration by parts. Although a significant number of those candidates who obtained the correct values for the two areas went on to subtract these values to obtain the correct final answer, some others added their correct values and so failed to gain the final two marks.

Q7.

This question on polar coordinates proved to be the most demanding question on the paper. In part (a), the better candidates had no problems in finding the cartesian equation for C_1 and rearranging it by completing the squares so as to be able to deduce that the curve was a circle. Many other candidates failed to eliminate θ even after a page or more of working and abandoned part (a) of the question.

Many candidates scored heavily on the more familiar part (b)(i), finding the area bounded by the second curve C_2.

Proving that the two curves did not intersect in part (b)(ii) was a challenge, even for the better candidates, and it was rare to award all four marks for this part of the question. However, some excellent solutions were seen which used a variety of methods including forming and solving quadratic equations in either $\sin x$, $\cos x$, $\tan x$, $\sec x$, use of calculus and use of the R, α form.

In the final part of the question, those who used part (a)(ii) to find the area enclosed by C_1 and used it with their answer to part (b)(i) had no problems scoring both marks. Those who tried to use integration to find the area enclosed by C_1 did not analyse the problem fully and failed to score. A common misinterpretation of the word 'intersect' is indicated by the not uncommon answer 'Since C_1 and C_2 do not intersect the required area is just 16.5π, the area bounded by C_2.'

Q8.

Most candidates were able, in part (a), to verify that the point with polar coordinates $(2, \pi)$ lay on the curve C. In part (b)(i), most candidates found the polar coordinates for the two points of intersection, although candidates should continue to be encouraged to give values of θ (in this case for which $\cos\theta = -\frac{1}{2}$) in the given range (in this case $0 \le \theta \le 2\pi$).

Although some excellent solutions were seen in part (b)(ii), in general candidates found this part to be a challenge. Many candidates gave a final answer which matched the area of triangle OLM. Without any clear identification of which area was being calculated, the examiner had to assume that a candidate's final expression was the candidate's answer for the area of triangle LMN.

Part (c), which required candidates to change a polar equation into a cartesian equation, caused more problems for candidates than anticipated. Those candidates who started by squaring both sides of the polar equation rarely scored more than one mark. In comparison, those candidates who rearranged to obtain $3r = 2 - 2x$ before squaring, usually went on to find a correct cartesian equation of the curve.

Q9.

This question, which tested the relationship between cartesian and polar coordinates, caused candidates more problems than anticipated. In part (a), it was not uncommon to see solutions which assumed incorrectly that the angle between OA and the x-axis was 45^o, which led to the incorrect coordinates ($\sqrt{50}, \sqrt{50}$) for A. In part (b)(i), a common wrong value for k was 5 but the value of tan α was usually stated correctly. Many candidates gave the correct polar equation for the circle in part (b)(ii).

Q10.

Those candidates who replaced sin 2θ by 2sin θ cos θ generally obtained the correct cartesian equation in part (a). The sketch of the curve C (rectangular hyperbola) required in part (b) was not answered as well as expected with many sketches consisting of closed loops. Candidates presented a variety of acceptable methods for part (c). Those who eliminated r were required to obtain a trigonometrical equation in a single angle before any mark was awarded. Usually candidates who had found the correct equation went on to obtain the correct exact values for the polar coordinates of A. The second most popular method involved working with the cartesian form of the equation of C. Candidates who recognised the cartesian form of the equation of the line equivalent to the polar form given in part (c) of the question generally had no difficulty getting the cartesian coordinates for A as (2, 2), but then a significant minority could make no further progress.

Q11.

Part (a) was generally answered correctly, with most candidates showing sufficient detail in reaching the printed result. Although most candidates started their solution for part (b) by successfully substituting $r\cos\theta$ for x and $r\sin\theta$ for y into either the given equation or the alternative form given in part (a), many failed to go on to score full marks because they did not consider and eliminate the negative square root (or the second solution of the quadratic equation).

Q12.

Although many candidates scored marks for using $r \sin \theta = y$ and $r = \sqrt{x^2 + y^2}$, a significant number failed to eliminate r correctly. Those who rearranged the equation to $r = 4 + y$ before squaring generally went on to present a fully correct solution.

Q13.

A very high proportion of candidates showed knowledge of the two trigonometric identities required in part (a). Part (b) was also answered correctly by many candidates although the errors '$x^2 + y^2 = r$' and '$\left(r^2\right)^3 = r^5$' were not uncommon. Most candidates scored at least two of the three marks in part (c)(i), with some losing the final mark for either the wrong value $\frac{\pi}{4}$ or for giving their second value outside of the stated domain.

In the final part of the question, it was pleasing to find most candidates stating the general formula for the area of a sector in polar coordinates (as given in the AQA formulae booklet) and then substituting and expanding for r^2 correctly. It was pleasing to see a large number of candidates write $\sin^2 2\theta$ correctly in terms of $\cos 4\theta$ although, for others, all the usual errors were presented, many of which led to the 'correct' value $\frac{3\pi}{4}$ for the area of a loop. Although many candidates correctly used their values obtained in part (c)(i) for the limits of integration, some used different values without any explanation or justification. A very small number of candidates did not show their method of integration in reaching the answer $\frac{3\pi}{4}$. Such candidates lost the same significant number of marks as those who, for example, used the wrong identity '$\sin^2 \theta = \frac{1}{2}(1 - \sin 4\theta)$' to integrate $\frac{1}{2}(1 + 2\sin 2\theta + \sin^2 \theta)$ and obtained the answer $\frac{3\pi}{4}$.

Q14.

Those candidates who used $y = r \sin \theta$ and $x^2 + y^2 = r^2$ had no problem reaching the printed answer in part (a). In part (b), although almost all candidates started with a correct integral for the area bounded by the curve, errors in the squaring of $2 \sin\theta + 5$ were not uncommon. The method for integrating $\sin^2\theta$ was well understood but sign errors in the subsequent integration were seen. In the final part of the question, many candidates applied a correct method to find the points of intersection P and Q but a significant number could not present a valid method to find the area of the quadrilateral.

Q15.

In part (a), some candidates started with $\frac{1}{2}\int_0^{2\pi} 6(1 - \cos\theta)\,d\theta$ and, without seeing a previous general formula, $\frac{1}{2}\int r^2\,d\theta$, examiners were unable to give the method mark. Although many candidates presented a correct solution for the area of the region bounded by the curve, errors in the squaring of $6(1- \cos\theta)$ were quite common. The method for integrating $\cos^2\theta$ was well understood and there was a lower proportion of sign errors in the subsequent integration than in the January 2006 exam. Although part (b) proved to be more of a challenge there were better than expected responses to both parts. The most common error was the use of '$r = 9$' rather than the correct value $r = 3$. Also in part (i) some candidates gave the polar coordinates of the two points in the wrong form; basically (x, y). The most successful approach for finding the length of AB was to use $AB = 2r\sin\theta$ ($= 2y$) although other correct methods included use of the cosine rule. Some candidates incorrectly assumed that triangle AOB was right-angled at O.